

Homework 8

For Monday, October 12, 2026.

Give exact answers unless an approximation is requested, and justify your reasoning. Do not use electronic tools. All functions are real-valued, and angles are measured in radians. Use the results covered through Section 8 of Lecture 9. For local extrema, consider only interior points of the stated domain. Use the convention $\operatorname{arcsec} x = \arccos(1/x)$, with values in $[0, \pi] \setminus \{\pi/2\}$.

1. Let $F(x) = \ln \left| \frac{x-1}{x+1} \right|$. Find the domains of F and F' , and compute F' .
2. Find the equation of the tangent to $y = \operatorname{arcsec}(3x)$ at $x = -2/3$.
3. Differentiate the following functions on $x > 0$:

$$(a) x^{\sqrt{2}}, \quad (b) (\sqrt{2})^x, \quad (c) x^x.$$

In each case, identify whether the base, the exponent, or both depend on x .

4. Let $f(x) = x(x+1)^{\sqrt{2}}$, for $x > -1$.
 - (a) Use logarithmic differentiation to find $f'(x)$ for $x \neq 0$.
 - (b) Find $f'(0)$ directly from the derivative definition. Explain why $\ln |f(x)|$ cannot be used at $x = 0$.
5. Evaluate the limit using derivatives and known limits:

$$\lim_{x \rightarrow 0} \frac{\arccos x - \pi/2}{\sin x}.$$

6. Suppose u, v are differentiable near 0, with

$$u(0) = 2, \quad u'(0) = -1, \quad v(0) = 3, \quad v'(0) = 2.$$

For $Y(x) = u(x)^{v(x)}$, explain why Y is defined and differentiable near 0, and find $Y'(0)$.

7. Find the linearization of $f(x) = \sqrt[3]{x}$ at $a = 8$. Use it to estimate $\sqrt[3]{8.12}$.

8. The linearization of a differentiable function f at $a = 2$ is

$$L_f(x) = 5 - 3(x - 2).$$

Find the linearization of $g(x) = \ln f(x)$ at $a = 2$. Justify that g is defined near 2.

9. Suppose f is defined near 1, is continuous at 1, and

$$\lim_{h \rightarrow 0} \frac{f(1+h) - 4}{h} = -2.$$

Determine $f(1)$, prove that $f'(1)$ exists, and find the linearization of f at 1.

10. Let $f(x) = 2 - x + |x|^{3/2}$.

(a) Use the derivative definition to find $f'(0)$, and find the linearization L at 0.

(b) Given $\varepsilon > 0$, find an explicit $\delta > 0$ such that

$$0 < |x| < \delta \implies |f(x) - L(x)| < \varepsilon|x|.$$

11. For $f(x) = 1/x$ at $a = 2$, let $dx = h$, where $|h| < 1$. Find dy , the exact change Δy , and $\Delta y - dy$. Which is larger, dy or Δy , when $0 < |h| < 1$?
12. Let f be continuous and strictly increasing on an open interval containing 1, and differentiable at 1, with $f(1) = 2$ and $f'(1) = 4$. Let $g = f^{-1}$. Find the linearization of g at 2, and use it to estimate $g(2.04)$, assuming 2.04 belongs to the domain of g .
13. Prove that there are no real constants A, B for which

$$\lim_{h \rightarrow 0} \frac{|h| - A - Bh}{|h|} = 0.$$

14. Suppose, for all sufficiently small h , including $h = 0$,

$$f(a+h) = A + Bh + h\rho(h), \quad g(a+h) = C + Dh + h\sigma(h),$$

where $\rho(h) \rightarrow 0$ and $\sigma(h) \rightarrow 0$ as $h \rightarrow 0$. Find a function $\tau(h)$, with $\tau(h) \rightarrow 0$, such that

$$f(a+h)g(a+h) = AC + (AD + BC)h + h\tau(h).$$

Hence identify the linearization of fg at a .

15. Find the absolute maximum and minimum values of $f(x) = 4 - (x - 1)^2$, and all points where they are attained, on each of the domains

$$(a) [-1, 2], \quad (b) (-1, 2).$$

If an extremum does not exist, justify this. Use algebraic inequalities.

16. Sketch by hand the graph of $f(x) = ||x| - 1|$ on $[-2, 2]$. Find all local and absolute maxima and minima, giving their values and all points where they are attained. Justify your conclusions from the graph or a piecewise-linear formula.

17. For each function and domain, decide whether the Extreme Value Theorem guarantees that both an absolute maximum and an absolute minimum are attained. Check the hypotheses explicitly.

(a) $f(x) = |x|$, $[-1, 1]$,

(b) $g(x) = \ln x$, $(0, 1]$,

(c) $k(x) = 1/(1 + x^2)$, $\mathbb{R}$.

18. For a real parameter c , define on $[0, 1]$

$$f_c(x) = \begin{cases} x, & 0 \leq x < 1, \\ c, & x = 1. \end{cases}$$

(a) For which c is f_c continuous on $[0, 1]$?

(b) For which c does f_c attain an absolute maximum? Give the value and its location whenever it exists.

(c) Determine the absolute minimum value and all its locations for every c .

19. Let f be defined on an open interval I , and let $c \in I$. Write the negation of the statement that f has a local maximum at c , using quantifiers and inequalities. Use this negation to show that $f(x) = x$, on $\mathbb{R}$, has no local maximum at any point.

20. Let f be continuous on $[a, b]$, where $a < b$, and suppose $f(x) \neq 0$ for every $x \in [a, b]$.

(a) Prove that there is a number $m > 0$ such that $|f(x)| \geq m$ for every $x \in [a, b]$, and deduce that $1/f$ is bounded.

(b) Does the same conclusion hold for every continuous nonzero function on $(0, 1)$? Prove it or give a counterexample.

Hint for (a): apply the Extreme Value Theorem to $|f|$.